

In the Eckart method, the change in potential energy along the minimum energy path is described by the following *Eckart function*⁶⁰:

$$V(x) = \frac{Ay}{(1+y)^2} + \frac{By}{(1+y)} \quad \text{where } y = e^{x/\Delta} \quad (13.28)$$

To ensure that the function passes through the reactants, products, and transition structures, the parameters A and B are defined as the following functions of the forward (V_f) and reverse (V_r) reaction barriers (where the reaction is taken in the exothermic direction):

$$A = (\sqrt{V_f} + \sqrt{V_r})^2 \quad \text{and} \quad B = V_f - V_r \quad (13.29)$$

The remaining parameter Δ is chosen so as to give the most appropriate fit to the minimum-energy path. If this fit is biased toward the points near the transition structure (where tunneling is most important), it can be calculated as the following function of the imaginary frequency $\nu_{\ddagger}$ (where c is the speed of light)^{47,49}:

$$\Delta = \frac{i}{2\pi c\nu_{\ddagger}} \sqrt{\frac{1}{8} \frac{(B^2 - A^2)^2}{A^3}} \quad (13.30)$$

The Δ value obtained from this expression is in mass-weighted coordinates, which enables the reduced mass to be dropped from the standard⁶⁰ Eckart formulas,⁴⁹ resulting in the following expression for the permeability of the reaction barrier $G(W)$ as a function of the energy W :

$$G(W) = 1 - \frac{\cosh(\alpha - \beta) + \cosh \delta}{\cosh(\alpha + \beta) + \cosh \delta}$$

where

$$\alpha = \frac{4\Delta\pi^2}{h} \sqrt{2W} \quad \beta = \frac{4\Delta\pi^2}{h} \sqrt{2(W - B)} \quad \delta = \frac{4\Delta\pi^2}{h} \sqrt{2A - \frac{h^2}{16\pi^2\Delta^2}} \quad (13.31)$$

The Eckart tunneling correction (κ) is then obtained by numerically integrating $G(W)$ over a Boltzmann distribution of energies, via⁵⁹

$$\kappa = \frac{\exp(V_f/k_B T)}{k_B T} \int_0^\infty G(W) \exp\left(-\frac{W}{k_B T}\right) dW \quad (13.32)$$

Although this expression requires numerical integration, it does not require sophisticated software and can be implemented easily on a spreadsheet; an Excel file for performing the calculation, given the forward and reverse barriers, the temperature, and the imaginary frequency, is freely provided at <http://rsc.anu.edu.au/~cylin/scripts.html>.

13.4.4 Beyond the Harmonic Oscillator Approximation

In most quantum chemistry packages, Eqs. (13.9) to (13.23) are used for calculating the partition functions and associated thermodynamic data, and thus assume an ideal gas under the rigid rotor/harmonic oscillator approximation. Under the harmonic oscillator approximation, the potential for each vibrational mode is treated as a parabolic well. For bond-stretching modes, this is a close approximation to the true potential energy surface; however, for a torsional mode, the potential energy surface may have multiple minima and is better described as a hindered internal rotation. For high-frequency modes ($\nu > 300 \text{ cm}^{-1}$), the contribution of these motions to the overall partition function is negligible at room temperature (i.e., $Q_{\text{vib},i} \approx 1$), and thus the error incurred in treating these modes as harmonic oscillators is not significant. However, for the low-frequency torsional modes, these errors can be significant and a more rigorous treatment is often necessary; this is especially the case for the reactions of relevance to free-radical polymerization.^{8,11,12b,c,l}

Ideally, one should solve the Schrödinger equation for the full multidimensional potential energy surface representing all active modes of a molecule, and use the resulting energy levels in Eq. (13.7) to obtain the partition functions; however, this is impractical for larger molecules. Instead, the approach that is usually adopted is to apply the harmonic oscillator approximation to all $3N - 6$ internal modes of a molecule (as in the standard formulas above), but then multiply the resulting vibrational partition function by a correction factor for each internal hindered rotor partition function. This factor is calculated as the ratio of the 1D-HR partition function to the corresponding “pure” vibrational partition function, as calculated from the second derivative of the rotational potential at the minimum-energy structure. Using approximations such as this, the 1D-HR model has been shown to provide reasonable results in situations where testing against more sophisticated treatments is possible.⁶¹

To obtain the 1D-HR partition function for any given low-frequency torsional mode, we first need to compile the full rotational potential $V(\theta)$ for the mode in question; studies have shown that a resolution of 60° is sufficient for accurate results.⁶² The potential should be compiled as a relaxed scan (i.e., at each dihedral angle, the dihedral angle is frozen but the rest of the molecule is fully optimized) and, as in ordinary geometry optimizations, low levels of theory, such as B3-LYP/6-31G(d) are usually sufficiently accurate. Having obtained the potential, this is then used to solve the one-dimensional Schrödinger equation for a rigid rotor:

$$-\frac{\hbar^2}{2I_r} \frac{d^2\Psi}{d\theta^2} + V(\theta)\Psi = \varepsilon_i\Psi \quad (13.33)$$

In this equation Ψ is the wavefunction, ε is the energy, I_r is the reduced moment of inertia, and $V(\theta)$ is the rotational potential, which for this purpose should be supplied at a high resolution. To this end, the 60° resolution potential is fitted with a Fourier series of up to 18 terms and then reevaluated at a resolution of